

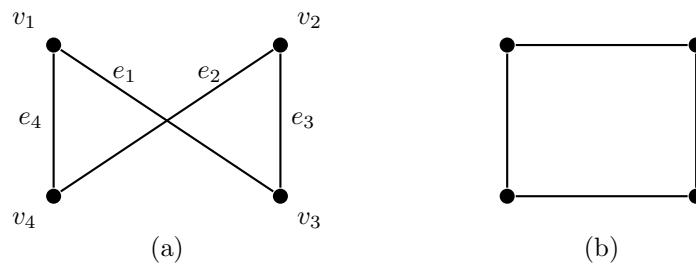
Weekly Assignment: Graphs and the Handshake Theorem (Epp 1.4 & 4.9)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 1.4 #4). Draw a picture of graph H , which has vertex set $\{v_1, v_2, v_3, v_4, v_5\}$ and edge set $\{e_1, e_2, e_3, e_4\}$ with the following edge-endpoint function:

Edge	Endpoints
e_1	$\{v_1\}$
e_2	$\{v_2, v_3\}$
e_3	$\{v_2, v_3\}$
e_4	$\{v_1, v_5\}$

Problem 2 (Epp 1.4 #6). Show that the two drawings represent the same graph by labeling the vertices and edges of drawing (b) to correspond to those of drawing (a).



Problem 3 (Epp 4.9 #8). Either draw a graph with four vertices of degrees 1, 2, 3, and 4, or explain why no such graph exists.

Problem 4 (Epp 4.9 #13). Either draw a simple graph with nine edges and all vertices of degree 3, or explain why no such graph exists.

Problem 5 (Epp 4.9 #15). A small social network contains three people who are network friends with six other people in the network, one person who is network friends with five other people in the network, and five people who are network friends with four other people in the network. The rest are network friends with three other people in the network. The network contains 41 pairs of network friends.

- How many people are network friends with three other people in the network?
- How many people are in the network?

Problem 6 (Epp 4.9 #20).

- Draw K_6 , a complete graph on six vertices.
- Use the result of Example 4.9.9 to show that the number of edges of a simple graph with n vertices is less than or equal to $\frac{n(n-1)}{2}$.

Problem 7 (Puzzle). A couple *really* wants to have a baby girl. They decide to have children until they have a girl, but realistically they will give up if they have four consecutive boys. Assume each child is independently equally likely to be a boy or a girl.

What is the expected number of boys and the expected number of girls in their family?

Hint: List every possible outcome and its probability, then compute the expectations directly.